SocioTrack

Closed-Loop Social Media Engagement Agent Core

LangGraph StateGraph + ChromaDB In-Memory Vector Store + FastAPI + Next.js

Team Ownership:

Person 1 (Agent Core) | Person 2 (FastAPI & DB) | Person 3 (Frontend UI) | Person 4 (Supabase DDL & Seeding)

1. The Problem: The Enterprise AI Content Trap

Why existing generative AI social workflows fail to drive practitioner traction:

*** 1. Generic Marketing Slop:**

Unconstrained LLMs write hollow hype ("revolutionary paradigm shift") that damages brand authority.

*** 2. Amnesiac Publishing:**

Marketing stacks fail to retain memory of past winning posts, sentiment signals, or viral styles.

*** 3. Zero Voice Governance:**

Without deterministic evaluation, posts frequently violate tone guidelines, formatting rules, and dos/donts.

*** 4. Broken Feedback Loops:**

Engagement metrics (shares, replies, likes) sit in analytics silos without training the creation engine.

2. The Solution: SocioTrack Agentic Core

A deterministic LangGraph StateGraph pipeline enforcing strict memory and evaluation boundaries:

-> Node 1: retrieve_brand_memory:

Pulls top 5 past high-performing posts + real-time audience sentiment from vector store.

-> Node 2: draft_post_variations:

Generates 3 distinct variations: Format A (Data-first), Format B (Hot take), Format C (Tactical breakdown).

-> Node 3: evaluate_brand_voice:

Fast critique pass scoring drafts (0-10) against brand dos and donts, selecting the winning post.

-> Node 4: update_memory_with_performance:

Computes composite score: (Shares*3 + Replies*2 + Likes*1) / Impressions * 100. If > 5.0, vectorizes lesson.

3. Reinforcement Learning Formula

Prioritizing high-intent viral distribution over low-friction vanity views:

Composite Score = ((Shares*3) + (Replies*2) + (Likes*1)) / Impressions * 100

Gating Condition: If Composite Score > 5.0, embed & insert key lesson into vector store

*** Shares (Weight 3x):**
Highest-intent viral action. Technical leads forwarding architecture blueprints to engineering teams.

*** Replies (Weight 2x):**
Algorithmic catalyst. Contrarian perspectives sparking deep practitioner debates in comments.

*** Likes (Weight 1x):**
Passive baseline sentiment indicator.

*** Threshold (5.0):**
Strict barrier to prevent mediocre posts from contaminating the long-term memory store.

4. Team Ownership & Modular Architecture

Clear modular boundaries across backend, database, and frontend:

Role	Owner	Core Deliverables
Person 1: Agentic Core	Person 1	LangGraph StateGraph, 4 nodes, ChromaDB Vector Store, Pydantic State
Person 2: API & Routing	Person 2	FastAPI routes (/generate-drafts, /posts/approve, /simulator), SQLAlchemy ORM
Person 3: Frontend UI	Person 3	Next.js 14 App, 3-column layout, DraftQueue, MemoryDrawer, SimulatorBar
Person 4: Data & Pitch	Person 4	Supabase DDL schema.sql, seed_db.py, test suite, pitch deck & demo video